



CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

CSIT335-G5 TESTING AND QUALITY ASSURANCE

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Teknotes

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1. Introduction

1.1 Purpose

TekNotes is a centralized digital platform designed to solve the problem of students spending hours searching the internet for reliable study materials. The primary purpose of the system is to provide students with easy access to well-organized, course-specific notes for both major and minor subjects. By putting all resources in one place, TekNotes helps students save time, reduce distractions, and focus on studying.

1.2 Scope

The TekNotes system manages the gathering, saving, and distribution of academic learning resources, alongside fostering student collaboration. The scope of this document covers the functional behavior of the two core modules: uploading and bookmarking materials, and creating, joining, and participating in interactive Study Groups.

1.3 Purpose of the document

This Functional Specification Document provides detailed information on how the TekNotes platform will operate. It defines the exact requirements and system behaviors for the file upload, bookmarking, and study group features (including group chat and file sharing) to ensure the final product meets the students' needs.

1.4 Project Scope

In Scope The system will include the following functionalities:

- Allowing students to upload notes, presentations, and documents.
- Enabling students to bookmark favorite study materials to their personal profile for quick access.

- Enabling students to create and join Study Groups (password-protected).
- Real-time group chat and file sharing/downloading within specific Study Groups.
- Searching and filtering Study Groups by Course Code and Teacher.

Out of Scope The following features are not included in the current system implementation:

- Direct 1-on-1 private messaging between students (outside of groups).
- Teacher portal for grading or reviewing materials.

1.5 Scope of the document

This document covers the functional specifications for two core features of the TekNotes system: 1) The Upload and Bookmark Module and 2) The Downloads Module.

1.6 Terms/Acronyms and Definitions

Term/Acronym	Definition
TekNotes	The centralized digital platform for sharing and accessing learning resources.
Upload	The process of transferring a file from a student's device to the TekNotes system.
Bookmark	A feature that lets a user save a link to a specific file inside their personal profile for fast retrieval.
Study Group	A collaborative digital space where students can chat, share files, and collaborate.

2. System Actors

2.1 User Roles (Actor/s)

Role	Description
Student	Uploads files, bookmarks resources, and downloads materials. (Primary actor).

3. Functional Specifications

3.1 Feature 1: Upload and Add Bookmark Mod

3.1.1 Upload File

1. This process allows students to contribute to the platform by uploading their own study materials. There is no restriction on the total number of files a student can contribute, allowing for an unlimited accumulation of academic resources. The 25MB limit is designed to accommodate comprehensive academic resources like high-resolution PDFs and lengthy presentations.
2. Flow:
 - **Click "Upload"** – On the main dashboard, tap the button to initiate the file upload process.
 - **Enter Title** – Input the material's title. Special characters and punctuation are fully permitted. There is no maximum character limit, though the UI will truncate or wrap exceptionally long text.
 - **Enter Course/Description** – Describe the material. There is no maximum character limit.
 - **Enter Teacher Name** – Input the instructor's name. There is no maximum character limit.
 - **Select File** – Choose a valid document from the device. Supported formats strictly include **.pdf**, **.ppt**, **.doc**, and **.docx**. The file size can be up to and including exactly 25MB.
 - **Upload** – Click the upload button to save the file. The system processes all uploads as new, independent entries; if a duplicate file is uploaded, it is

assigned its own unique metadata record and database ID without triggering any block or warning.

A confirmation message displays after successful upload.

3.1.2: Add/Remove Bookmark

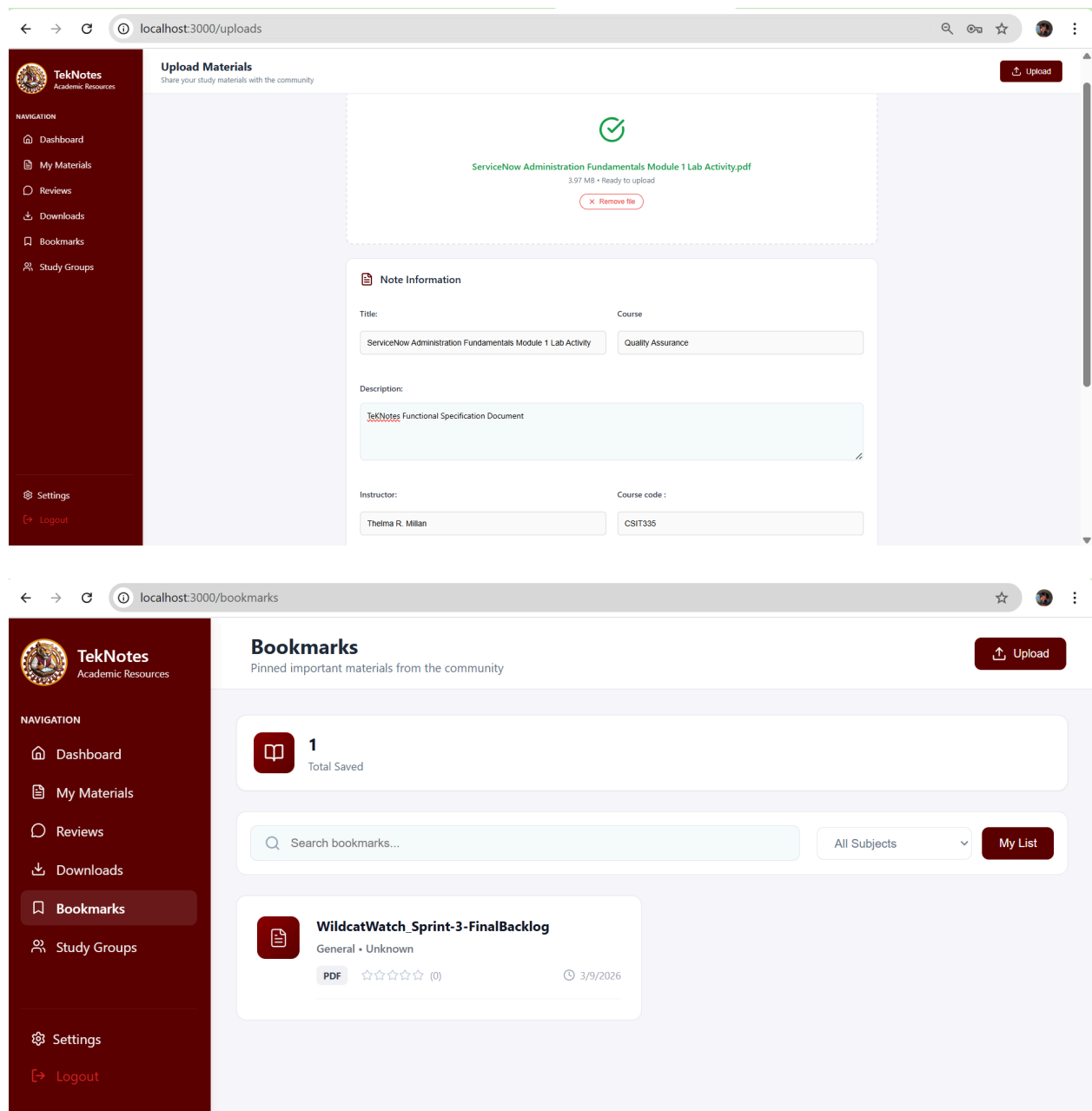
1. This feature lets users save a link to a specific file inside their personal profile. There is no limit to the number of bookmarks a student can save.
2. Flow
 - **Browse resources** – Student browses the repository and finds a useful file card.
 - **Click the “Bookmark” icon** – Tap the bookmark icon on the file card to save it. The system updates the list immediately without refreshing the page. There is no limit to the number of bookmarks a user can maintain.
 - **Toggle to Remove** – Users cannot bookmark the same file multiple times. If a file is already bookmarked, clicking the active Bookmark icon again will remove it from the "Saved Preferences" profile page.
 - **Handle Deleted Resources** – If an original uploader deletes a file that a user has bookmarked, the item remains in the bookmark list but will display a "File Unavailable" state.

3.1.3: Add/Remove Bookmark

1. This process allows students to manage the resources they have already uploaded to the platform, ensuring outdated or incorrect files can be removed.
2. Flow:
 - **Navigate to My Materials** – The student opens their personal dashboard containing their uploaded files.
 - **Select Edit** – The user updates the Title, Course, or Teacher fields and clicks save.
 - **Select Delete** – The user clicks the delete icon to remove the file from the database entirely.
 - **System Resolution** – If a file is deleted, the system automatically removes that file's reference from the bookmark lists of any other users who had saved it, replacing it with a "File Unavailable" state if accessed directly.

A confirmation message 'File updated successfully' or 'File deleted' displays.

3.1.3 Mock-up



3.2 Feature 2: Study Group Module

3.2.1 Purpose/ Description

This feature allows students to form collaborative communities. Students can search for existing groups using filters, create new groups

(password-protected), and interact within the group via a real-time chat interface and a dedicated "Shared Files" repository.

3.2.2 Use case

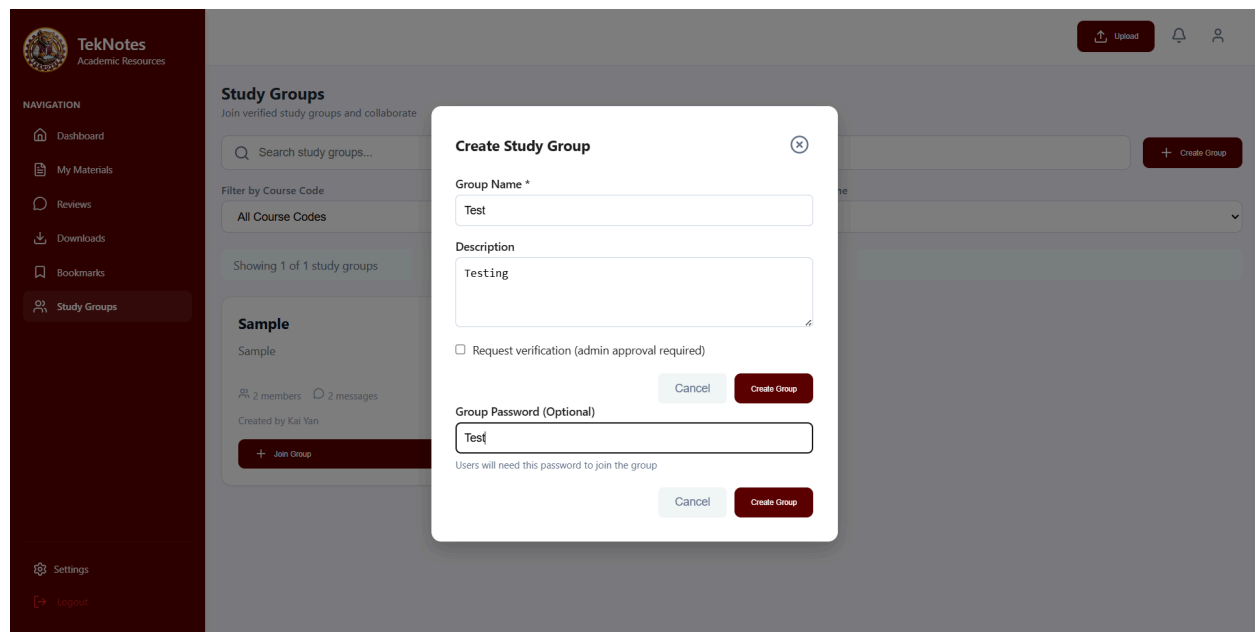
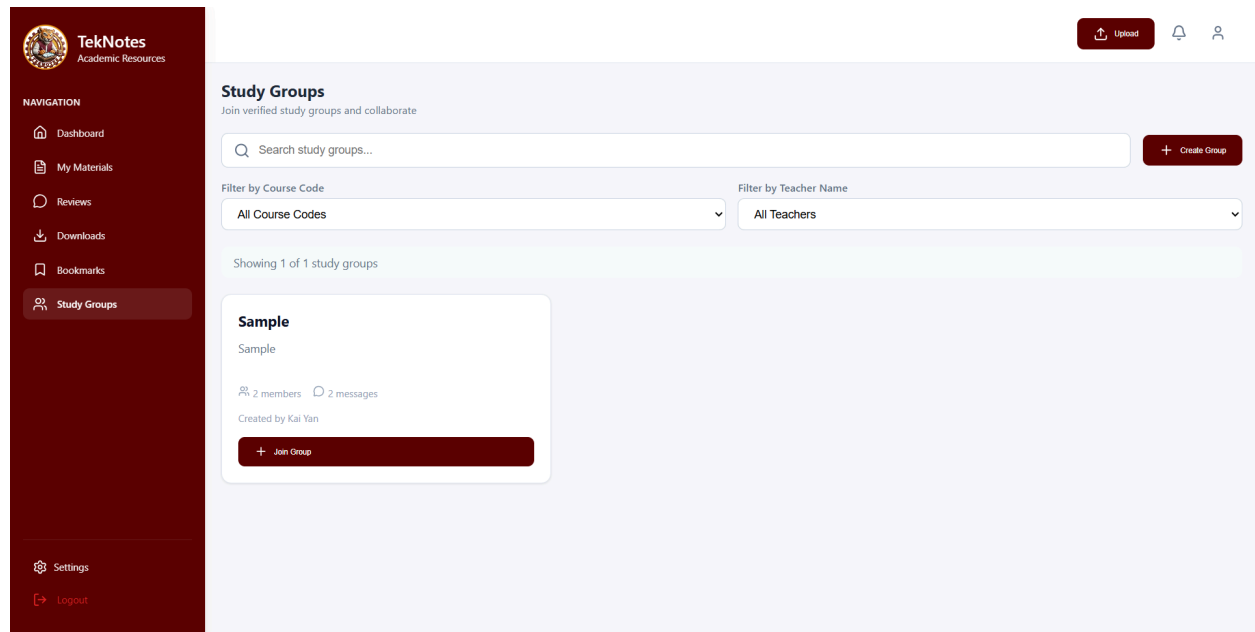
Use Case ID	UC-2
Use Case Name	Create, Join, and Interact in Study Group (Group Chat & File Sharing)
Primary Actor(s)	Student
Stakeholders and Interest	Students need to form secure, focused communities to discuss coursework in real-time and share specific files without outside distractions.
Trigger	Student navigates to the "Study Groups" tab to find/create a group, OR opens an active group to send a message/file.
Pre-conditions	Student is logged into the system. To chat or share files, the student must have successfully joined the password-protected Study Group.
Post-conditions	A new password-protected group is created, OR the student is granted access, OR the group chat history and shared files repository are updated with new content.
Main Success Scenario	1. Student uses the "Search" or "Filter" fields to find relevant groups. 2. To create a group, the student clicks "+ Create Group", manually inputs the Group Name, Description, and a custom Group Password, then clicks "Create Group". 3. System saves the user-defined

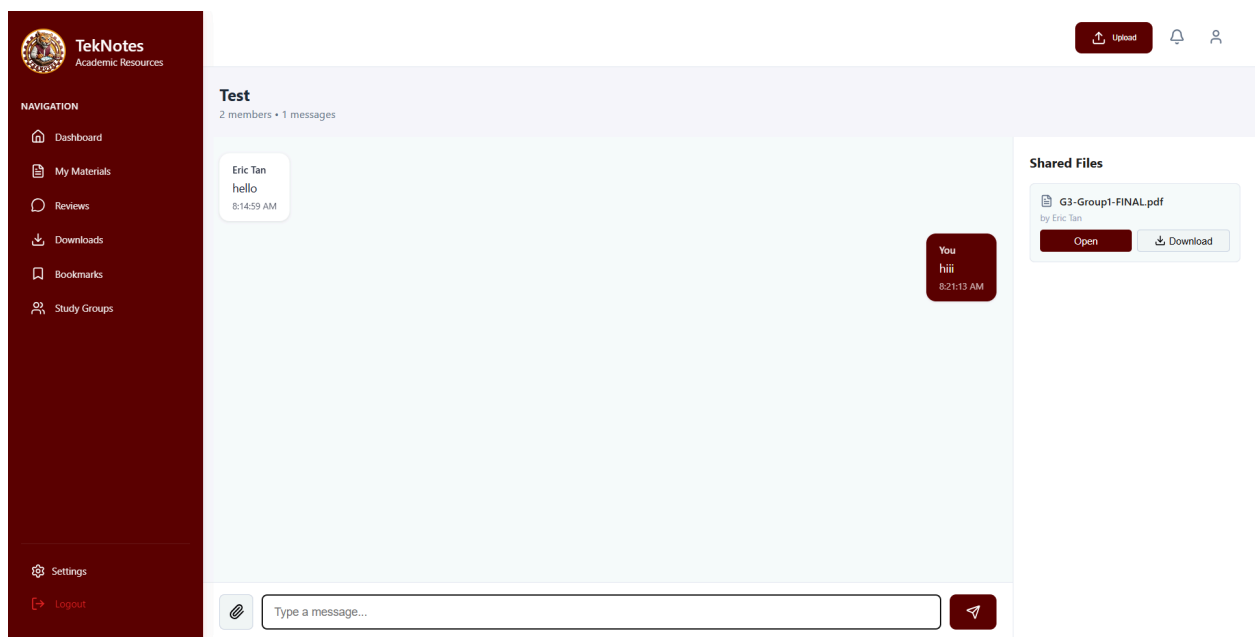
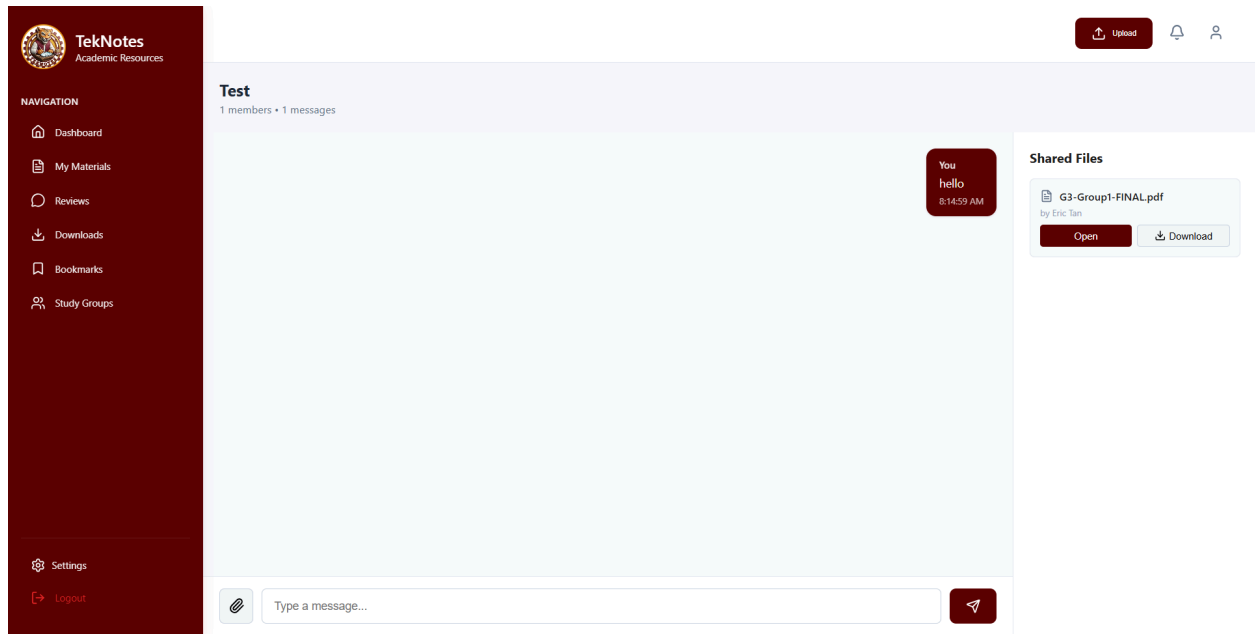
	password and generates the new group space. 4. To join an existing group, a student clicks "+ Join Group" on a group card. 5. System prompts the student for the group password. 6. Student manually inputs the correct password and submits. 7. System grants access and redirects the student to the group's chat interface. 8. Inside the group, the student types a message in the text box and clicks send. 9. System displays the message instantly with a timestamp. 10. Student clicks the paperclip icon to upload and share a file. 11. System displays the uploaded file in the "Shared Files" panel, allowing members to "Open" or "Download" it. 12. A student can join multiple study groups simultaneously. 13. A student can successfully leave a group they have previously joined. 14. The group creator completely deletes the group.
Extensions	2a. The student leaves the mandatory "Group Name" field blank. 1. System prevents the creation. 2. System displays a "Group Name is required" error message. 6a. The student enters an incorrect password.

	1. System denies access. 2. System displays an "Incorrect password. Please try again" error message. 8a. The student attempts to send a blank message. 1. System disables the send button or ignores the action. 2. No message is posted to the chat. 8b. The connection is lost while sending a message. 1. System drops the REST API request. 2. Message fails to send and is not saved to the database. 10.a The student attempts to attach a file that exceeds the maximum file size limit (e.g., 5MB). 1. System stops the upload process. 2. System displays a "File size exceeds limit" error message.
Priority	High
Special Requirements	The password input field must be masked (e.g., displaying bullets or asterisks) while the user is typing to ensure privacy. • Chat messages and new file uploads must appear instantly on the screens of all active group members without them needing to refresh the page.

- The chat interface must clearly distinguish the active user's messages from other members' messages visually (e.g., different alignment or bubble color).

3.2.3 Mock-up





3.2.4 Functional Requirements

Spec ID	Specification Description	Business Rules/Data Dependency
FR-SG-01	The system shall allow a student to create a new Study Group with a	Group Name is mandatory. System must save the manual

	custom name, description, and manual password.	password string exactly as inputted by the creator.
FR-SG-02	The system shall require students to input the correct password to join a protected Study Group.	Access is only granted if the user's input strictly matches the StudyGroup.password stored in the database.
FR-SG-03	The system shall facilitate text messaging within the group chat interface, allowing users to send, edit, and delete their own messages.	Messages must be tied directly to the StudyGroup.id and include the sender's details and a timestamp.
FR-SG-04	The system shall allow active group members to upload files of any format to, and download files from, the "Shared Files" panel. Any user in the group can delete a shared file.	Uploads are restricted by a maximum file size of 5MB. No malware scanning is implemented.

3.2.5 Field level specifications

Field Label	Validation / Business Rules	Error Messages	Data Dependencies
Group Name	Mandatory text field. Maximum 50 characters. No uniqueness requirement (multiple groups can share the	"Group Name is required."	Saved to the database as StudyGroup.name .

	same name).		
Group Password	Text field. Input must be masked. No minimum length; maximum 255 characters. No complexity requirements. Stored in plain text. No limit on incorrect attempts.	"Incorrect password. Please try again." (Triggered upon failed join attempt).	Compared against or saved as StudyGroup.password .
Chat Message Input	Text field. Database uses TEXT type supporting up to 65,535 characters. Cannot trigger send if blank.	N/A (Send button remains disabled/inactive)	Payload saved as ChatMessage.content . Depends on an active WebSocket or server connection.
Attach File (Paperclip)	File must not exceed 5MB. All file formats and extensions are accepted..	"File size exceeds limit."	Depends on server storage capacity. File path is linked to the specific StudyGroup.id .

4. Non-Functional Requirements

Performance:

- Uploading a standard document (under 5MB) within a Study Group or to the main repository must process in under 10 seconds on a standard internet connection.
- Uploading a file (under 5MB) within a Study Group uses a standard connection (2-5 Mbps upload).

- Real-time chat messages must appear in the UI with a latency of less than 500ms.
- Chat messaging and file sharing rely on REST APIs, requiring the client to refresh or poll the server to retrieve new content (messages do not appear instantly without polling).
- Navigating into a joined Study Group must reflect on the UI instantly (under 200ms).
- Bookmarking a file or navigating into a joined Study Group must reflect on the UI instantly (under 200ms).

Usability:

- The platform must feature a clean, minimalist interface to reduce student distraction and focus strictly on learning materials and group collaboration.
- The chat interface must clearly distinguish the active user's messages from other members' messages visually (e.g., different alignment or bubble color).
- Error messages for failed group creations, incorrect passwords, or failed uploads must be written in simple, non-technical language.

Security:

- All uploading, group participation, and chatting actions require the student to have an active, authenticated session.
- Access to a group's chat history and shared files must be strictly restricted to users who have successfully provided the group's specific password.
- Group passwords are stored in plain text in the database.
- There is no limit or lockout mechanism for incorrect password attempts.

Reliability & Availability:

- The centralized file repository, group directories, and real-time chat servers must have an uptime of 99% to ensure students can access their notes and collaborate during late-night study sessions.

Data Fields (Key Entities)

Entity / Field	Type	Required	Description
Resource.id	String	Yes	Auto-generated unique identifier for the uploaded file.
Resource.title	String	Yes	Name of the study material (e.g., "CSIT335 - QA").
Resource.courseDescription	String	Yes	Describe the context of the file uploaded (e.g., "Notes for Lesson 1-5").
Resource.teacher	String	Yes	The name of the teacher or professor associated with the material.
Resource.fileType	Enum	Yes	The format of the file (PDF, PPT, DOC).
Bookmark.id	String	Yes	Auto-generated identifier for the saved preference.
Bookmark.studentId	Text	Yes	Links the saved item to the specific student's profile.
Bookmark.resourceId	DateTime	Yes	Links to the specific file that was saved.
StudyGroup.id	String	Yes	Auto-generated unique identifier for the study group.
StudyGroup.name	String	Yes	The user-defined name of the group.
StudyGroup.description	String	No	A brief description of the group's purpose.
StudyGroup.password	String	Yes	The manual custom password set by the group creator.
ChatMessage.id	String	Yes	Auto-generated identifier for the message.

ChatMessage.groupId	String	Yes	Links the message to the specific study group.
ChatMessage.senderId	String	Yes	Links the message to the user who sent it.
ChatMessage.content	String	Yes	The text payload of the message.
ChatMessage.timestamp	DateTime	Yes	Date and time the message was sent.